

2. Processing, representing and analysing data		
(a) Tabulation, diagrams and representation		
2a.01	Represent data sets pictorially using calculated key values as necessary, and interpret and compare data sets displayed pictorially: tabulation, tally, pictogram, <u>pie chart</u> , <u>stem and leaf diagram</u> , <u>Venn diagram</u> .	Use of two-way tables is expected. Diagrams should have a key where appropriate. <u>Stem and leaf diagrams need to be ordered to allow identification of key values.</u>

Foundation tier content

What students need to learn:		Guidance
(a) Tabulation, diagrams and representation <i>continued</i>		
2a.02	Interpret and compare data sets displayed pictorially: <u>population pyramid, choropleth map</u> .	
2a.03	Represent data sets graphically using calculated key values as necessary, and interpret and compare data sets displayed graphically: bar charts, line graphs, time series, scatter diagrams, <u>bar line (vertical line) charts, frequency polygons, cumulative frequency (discrete and grouped) charts, histograms (equal class width), and box plots</u> .	Use of multiple and composite (including percentage composite) bar charts is expected. <u>No distinction will be made between cumulative frequency polygons (other than step polygons) and curves, while frequency polygons could be open or closed.</u> <u>Note: the 'y' axis of histograms may be labelled 'frequency' where equal class widths are used. (Frequency density is Higher tier only.)</u>
2a.04	(Higher tier topic only; not assessed at Foundation tier)	
2a.05	Justify the appropriate format and produce accurate visualisation of data.	Be familiar with the capabilities and advantages of using statistical software and spreadsheets to produce suitable diagrams and graphs, and know to avoid the inappropriate use of such technology. Appropriate format could take account of target audience. e.g. realising when a simple visualisation of data is appropriate, and when a more technical visualisation is appropriate.
2a.06	Recognise where errors in construction lead to graphical misrepresentation, including but not limited to incorrect scales, truncated axis, distorted sizing.	Correct use of class boundaries is required. Understand the possible distortion when interpreting 3D representations.
2a.07	Extract and calculate corresponding values in order to compare data sets that have been presented in different formats and be able to present the same information in multiple formats.	Including extracting information from spreadsheets, lists of statistics or graphs produced by statistical software.
2a.08	Select <u>and justify</u> appropriate form of representation <u>with regard to the nature of data</u> .	e.g. <u>scatter diagrams for bivariate data, histograms for grouped data, etc</u>

What students need to learn:		Guidance
(a) Tabulation, diagrams and representation <i>continued</i>		
2a.09	Determine skewness from data by inspection.	For example, know that positive skew could be indicated by: • mean > median > mode • median – LQ < UQ – median
2a.10	Interpret a distribution of data in terms of skewness identified from inspection.	For example, with positive skew know that values above the median have a greater spread than values below the median.